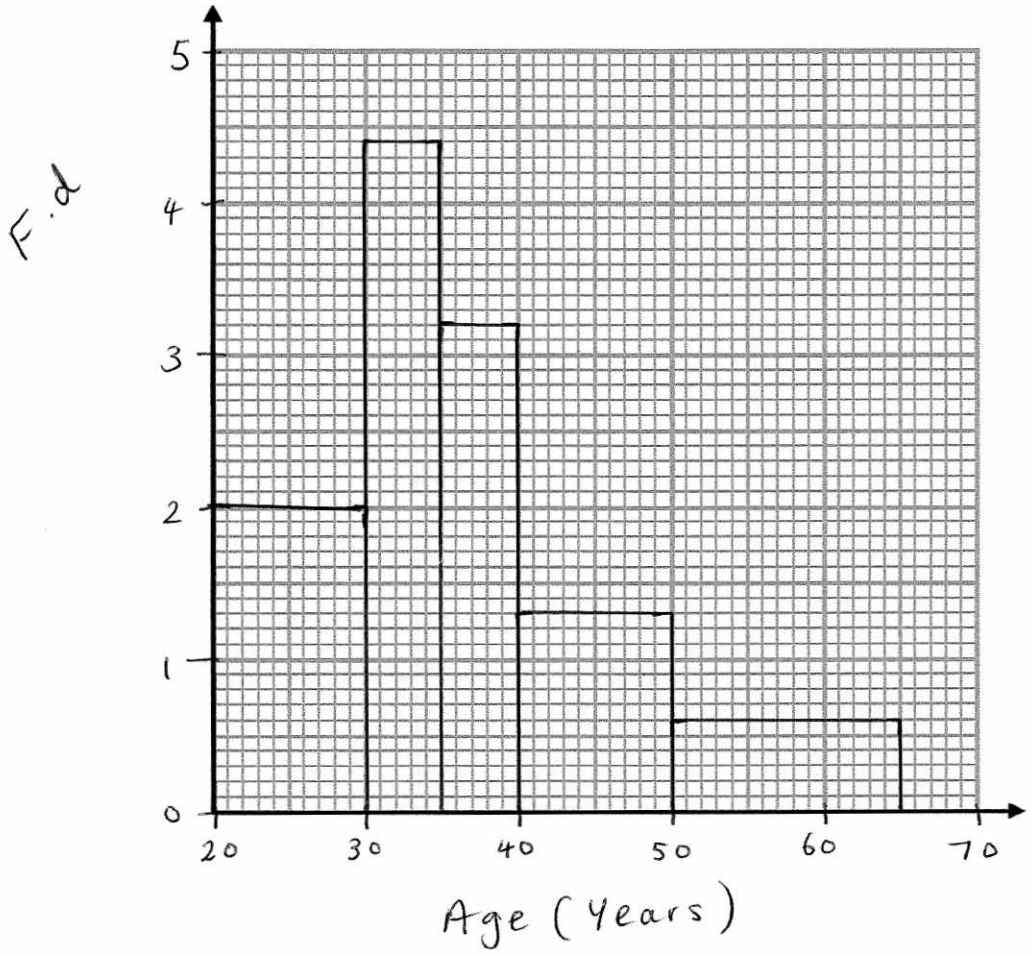
1 The table shows information about the age of 80 teachers.

$F.d = \frac{Freq}{width}$

Age (years)	Frequency	F.d.	
$20 < a \leqslant 30$ 10	20	2	[20 ÷ 10]
$30 < a \leqslant 35$ 5	22	4.4	[22 ÷ 5]
$35 < a \leqslant 40$ 5	16	3.2	[16 ÷ 5]
$40 < a \leqslant 50$ 10	13	1.3	[13 ÷ 10]
$50 < a \leqslant 65$ 15	9	0.6	[9 ÷ 15]

On the grid, draw a histogram for the information in the table.

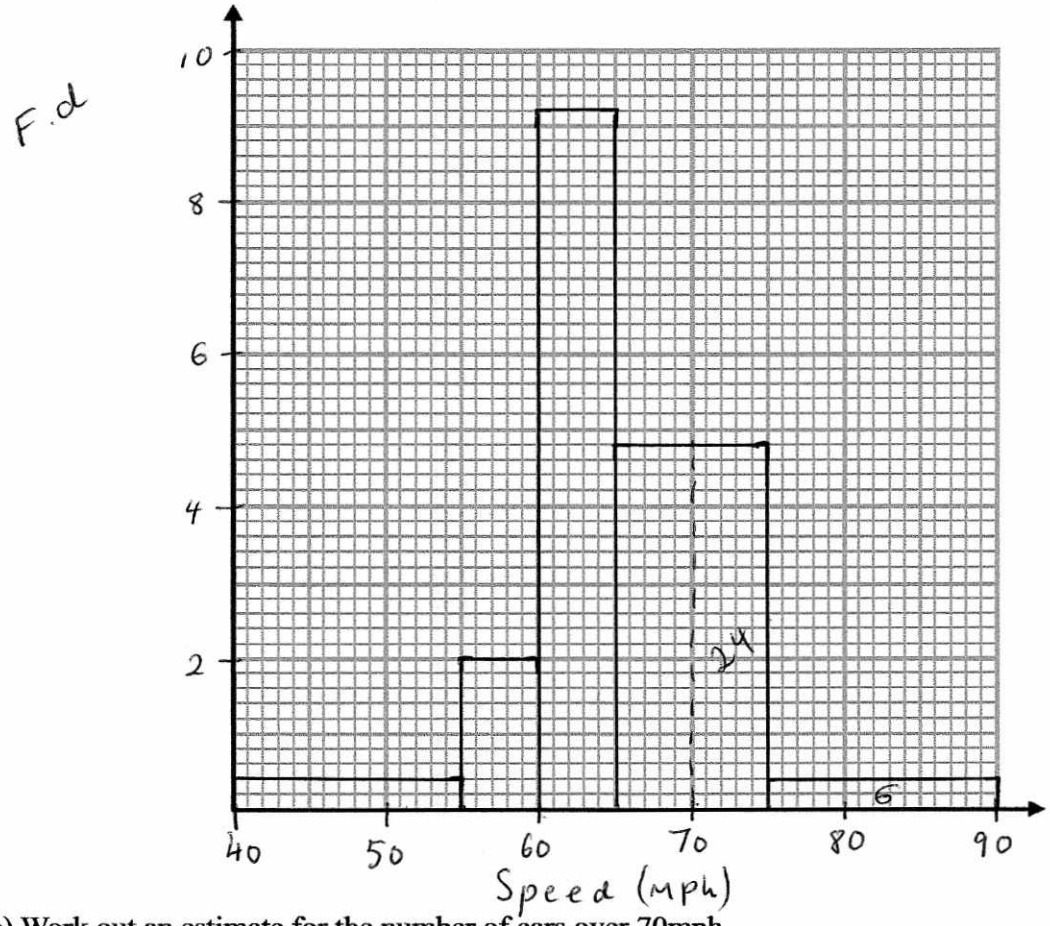


(Total for question 1 is 3 marks)

2 The table shows information about the speed, in mph, of some cars.

Speed (mph)	Frequency	F.d.
$40 < s \leqslant 55$ 15	6	0.4
$55 < s \leqslant 60$ 5	10	2
$60 < s \leqslant 65$ 5	46	9.2
$65 < s \leqslant 75$ 10	48	4.8
$75 < s \leqslant 90$ 15	6	0.4

(a) On the grid, draw a histogram for the information in the table.



(3)

(b) Work out an estimate for the number of cars over 70mph.

30

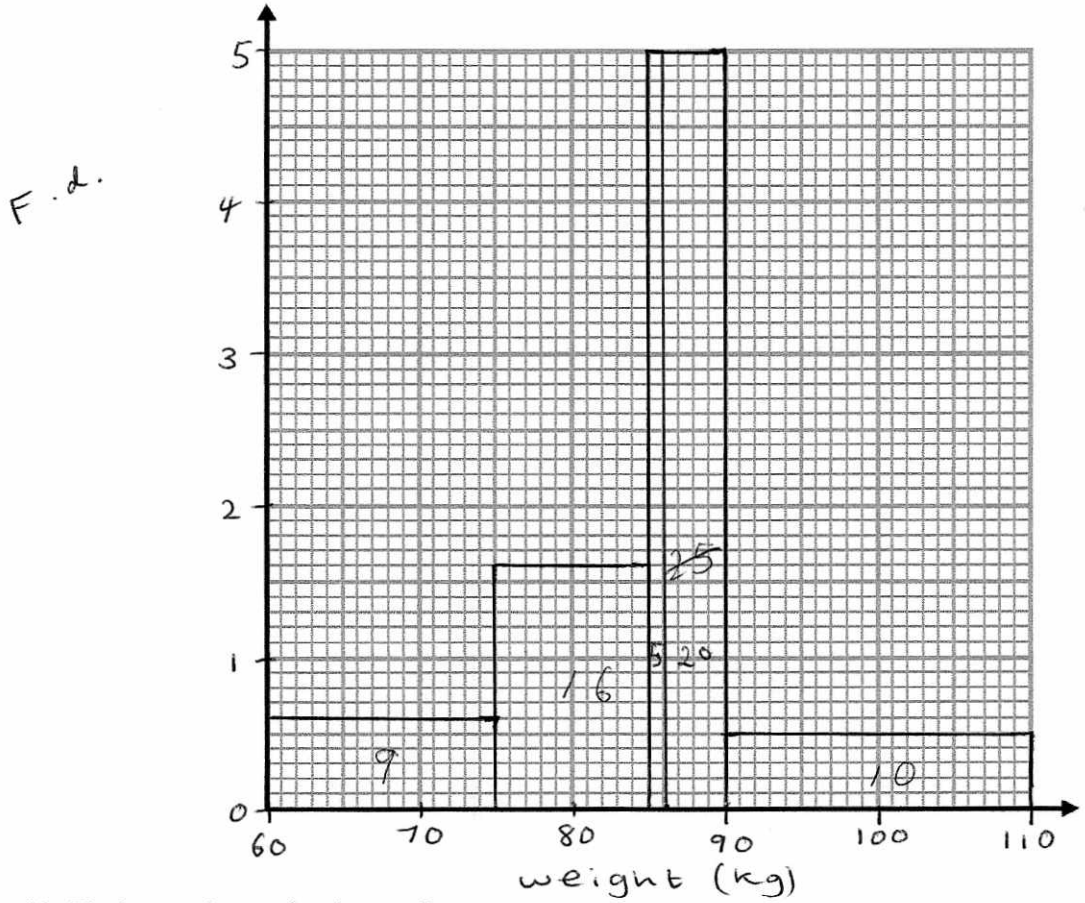
(1)

(Total for question 2 is 4 marks)

3 The table shows information about the weight of 60 pigs.

Weight (kg)	Frequency	F.d.
$60 < w \leqslant 75$ 15	9	0.6
$75 < w \leqslant 85$ 10	16	1.6
$85 < w \leqslant 90$ 5	25	5
$90 < w \leqslant 110$ 20	10	0.5

(a) On the grid, draw a histogram for the information in the table.



(3)

(b) Find an estimate for the median.

$\frac{60}{2} = 30$

30 on each side

86 kg

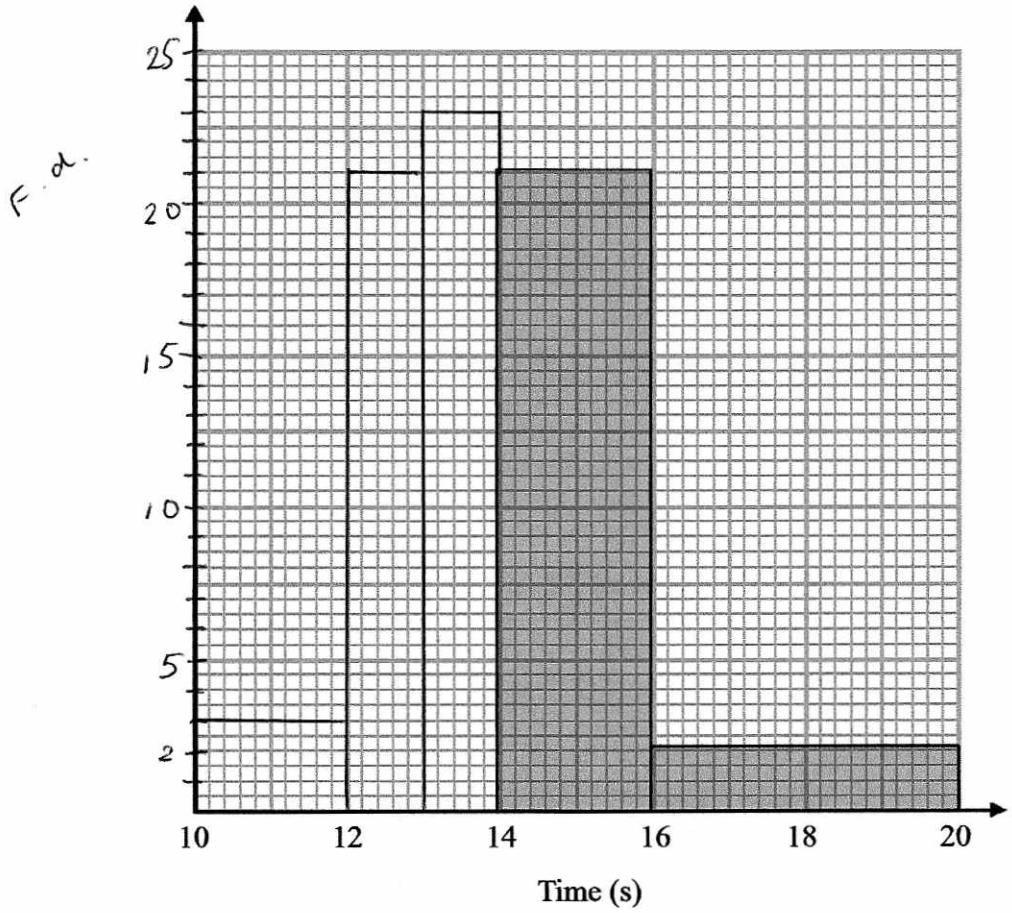
(2)

(Total for question 3 is 5 marks)

4 The table shows information about the time, in seconds, taken for some people to run a 100m race.

Time (s)	Frequency	F.d
$10 < t \leqslant 12$	6	3
$12 < t \leqslant 13$	21	21
$13 < t \leqslant 14$	23	23
$14 < t \leqslant 16$	42	21
$16 < t \leqslant 20$	8	2

(a) Use the information on the table to complete the histogram.



(b) Use the histogram to complete the table.

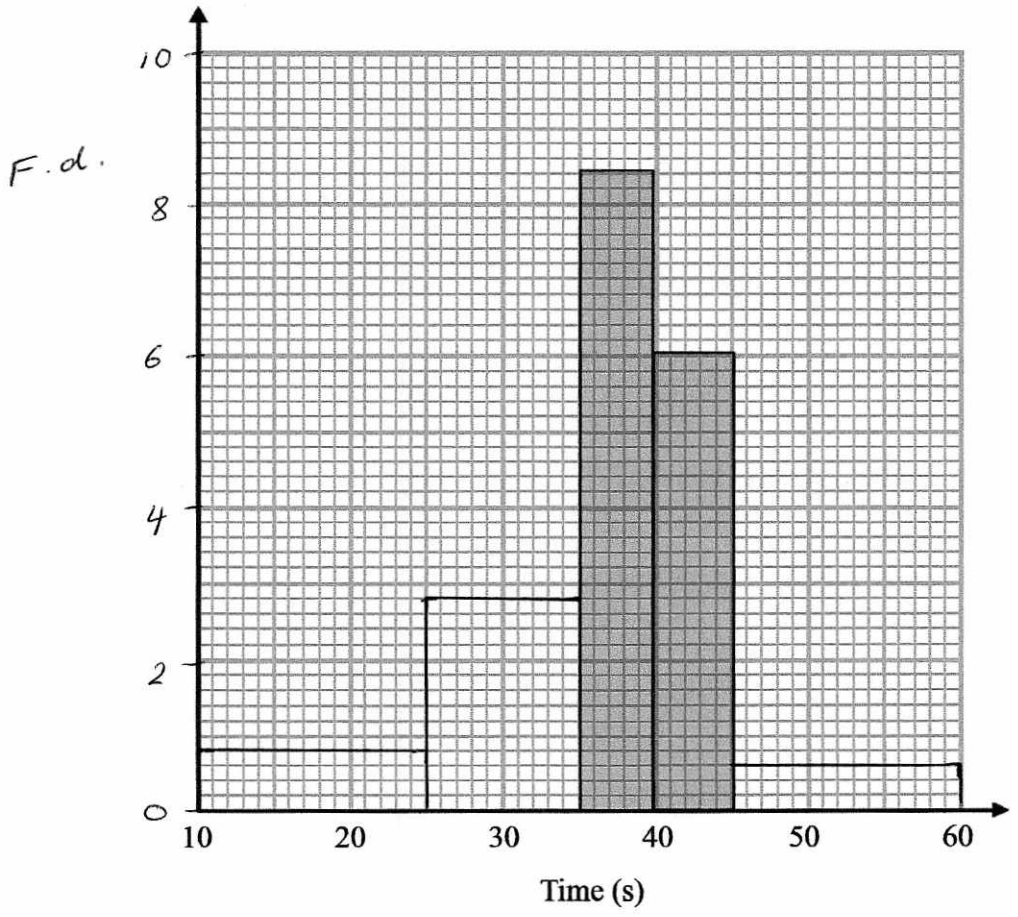
$21 \times 2 = 42$

(Total for question 4 is 4 marks)

5 The table shows information about the time, in seconds, taken for some people to complete a puzzle.

Time (s)	Frequency	F.d
$10 < t \leqslant 25$	12	0.8
$25 < t \leqslant 35$	28	2.8
$35 < t \leqslant 40$	42	8.4
$40 < t \leqslant 45$	30	6
$45 < t \leqslant 60$	9	0.6

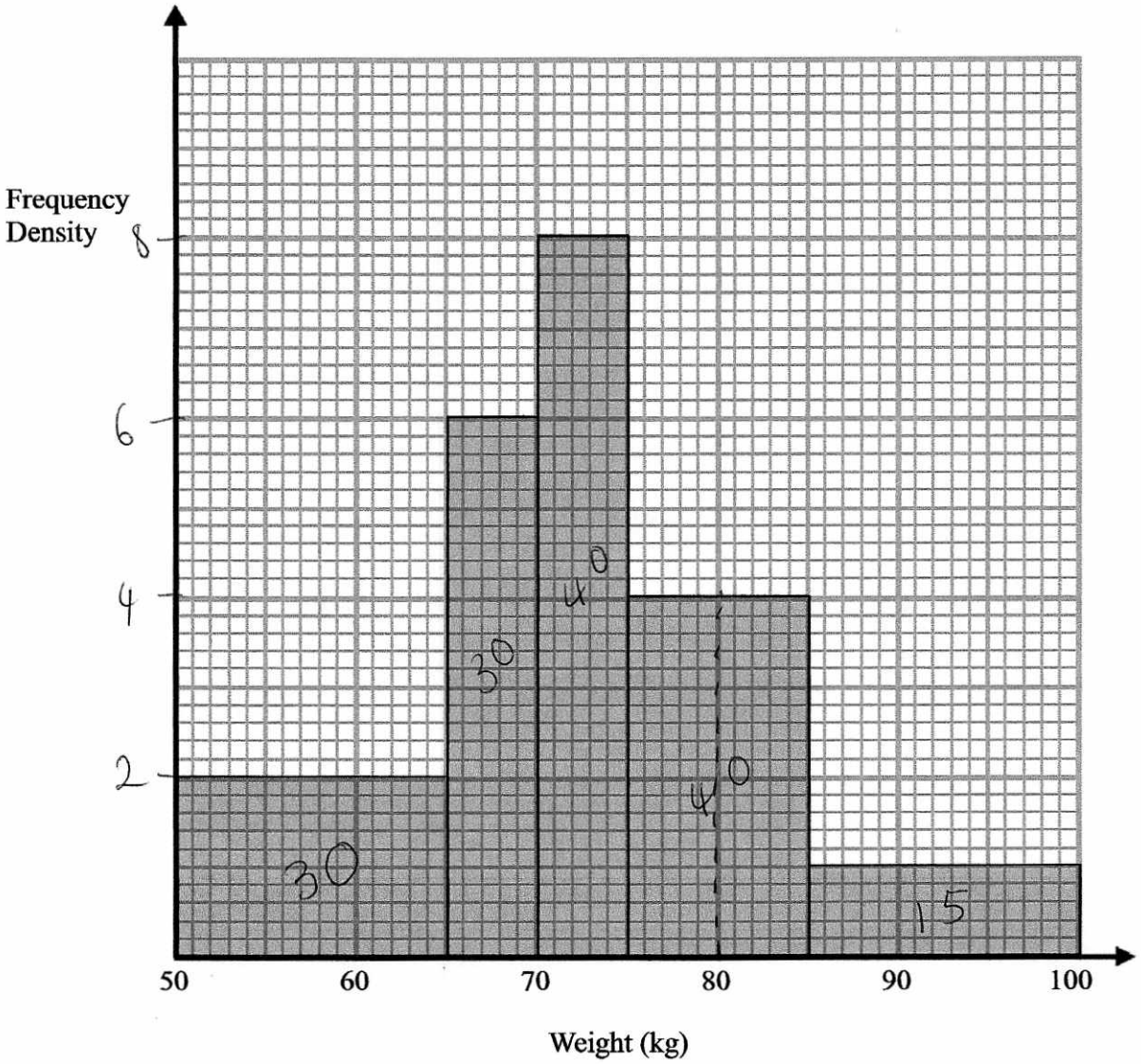
(a) Use the information on the table to complete the histogram.



(b) Use the histogram to complete the table.

(Total for question 5 is 4 marks)

6 The histogram shows information about the weight of pigs.



30 pigs weigh between 50 and 65 kg.

(a) Work out an estimate for the number of pigs which weigh more than 80kg.

20 + 15

35

(3)

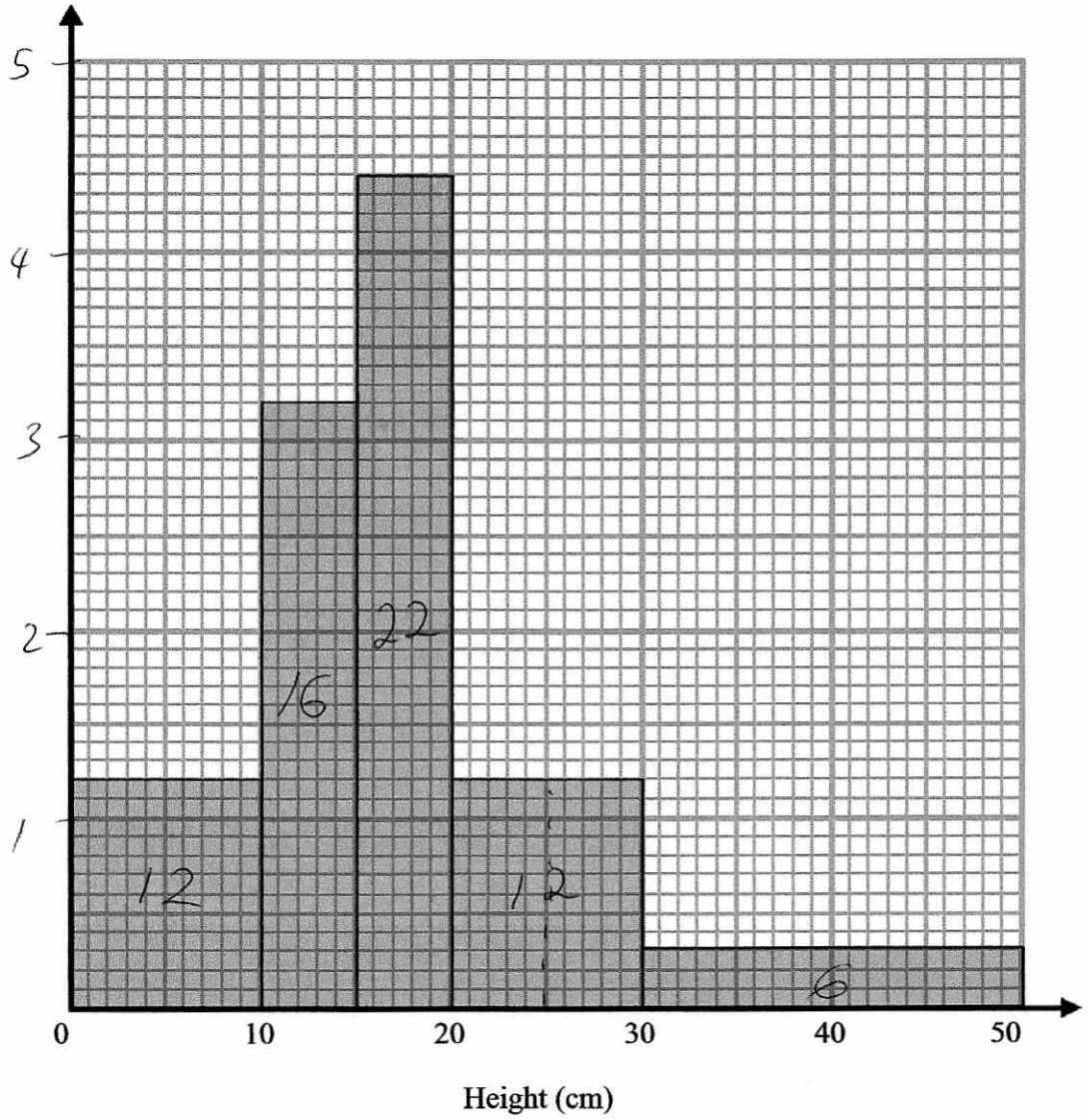
(b) Explain why your answer to part a is only an estimate.

we are using grouped data. we don't know how many pigs weigh between 80 and 85kg.

(1)

(Total for question 6 is 4 marks)

7 The histogram shows information about the height of some plants.



(a) Work out an estimate for the proportion of plants over 25cm tall.

$6 + 6 = 12$

$12 + 16 + 22 + 12 + 6 = 68$

$\frac{12}{68}$ or $\frac{3}{17}$

$\frac{12}{68}$

(3)

(b) Explain why your answer to part a is only an estimate.

we are using grouped data. we don't know how many plants are between 25 and 30 cm tall

(1)

(Total for question 7 is 4 marks)